The Batarang is a version of the Japanese Shuriken, or the throwing star

CINEMATECH

We have a closer look at the technology depicted in recent movies

The Dark Knight

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Cape

Batman's cape can be used to glide short distances. When used for short flights, the cape takes a hard-winged form. However, under normal circumstances, the cape hangs loosely from Batman's neck. Lucius Fox, CEO of Wayne Enterprises, describes the cape as being made of "memory cloth" that can take any form or structure around a specified skeleton. The fabric switches from a cape to crude wings on being exposed to a magnetic field. This concept is based on what adventure sports enthusiasts call a wingsuit. The wingsuit transforms the shape of an entire human body into a single aerofoil blade, or wing. This is accomplished by fabric stretching between the arms and the waist, and between the legs. Moreover, the wingsuit requires the use of a parachute for a safe landing. Batman doesn't use a parachute – an impossibility with available technologies. However, there are many materials such as the memory cloth used by Batman, available in the market. The mechanism mostly involves shape changing when exposed to heat. These materials can be deformed, and on heating return to their original shape. Another category of materials take on one form under high temperatures, and another at low temperatures. One of the practical applications of such materials would be a car that can never get dented, because heat returns it to its original form. Materials that change shape based on magnetism also exist, but are not as robust as the heat-based memory materials.

Tumbler

The tumbler was designed to be the hybrid between a sleek sports vehicle and an armoured vehicle. Since billionaire vigilantes are rare in the real world, a car such as the tumbler would possibly never exist. A man called Bob Dullam had built a fully working tumbler look-alike in his garage. However, that was just a shell without all the mechanisms inside. For one, the tumbler has a number of modes to frustrate, confuse or evade enemies. There are a whole range of weapons hidden under the hood. The front wheels have no axle, and are powered by independent engines. This means that the car can turn very sharp angles at very high speeds. Moreover, the batpod is modelled as a part of the tumbler. While there have been vehicles that could split into two and then rejoin, these have remained concept cars. What allows the tumbler such incredible speeds despite being heavily armoured is the aerodynamics. Similar to aeroplane wings that provide lift, the tumbler has smaller wings that provide drag. Air speed below the car is greater than above the car, causing a lowering of pressure which makes the tumbler stick to the ground. The same principle was used by the design teams of some F1 cars, but was later banned and has not been revived in the 2009 rules. The tumbler is powered by a jet engine. Some street cars with jet engines have been built, but the mileage made the idea very unpopular. Ignoring the urban legends, there have been some modders who have added an additional street-legal jet engine to their cars.

Suit

Batman's suit is made up of numerous small parts. This makes sure that Batman can move around smoothly. Kevlar is used in the suits. In fact, the suit itself is made up of biweave Kevlar, while some parts on the chest, and the limbs have extra armour. These parts are "hardened Kevlar plates over titanium-dipped tri-weave fibres". Kevlar, a fabric by DuPont, is used by security agencies and armies around the world. It is a kind of synthetic fabric that has a number of useful properties and is used from tyres for racing cars to the cones of headphones. Kevlar is heat resistant, and has a high tensile strength. When a bullet hits Batman's suit, even at close range, the fabric expands and changes shape to accommodate the bullet. The more the number of "weaves" in a Kevlar piece, the better is its resistance to bullets. Even Batman's utility belt is also made of Kevlar, with several gadgets including a small motor to power a grappling mechanism that allows Batman to ascend on a kevlar wire. Though there is no existing technology to pack so much power into such a small motor, there are a few devices in existence that allow people to grapple over wires for search-and-rescue operations. The Atlas rope ascender devised by an MIT student has a similar mechanism.